Search projection

Discover references.
Expose how fresh the answer is.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A library catalogue with a visible indexing date: “nothing matches” differs from “the catalogue is behind.”

Why it matters: Consumers must discover answers without receiving answer references. A replayable source and explicit watermark make that discovery inspectable and rebuildable.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

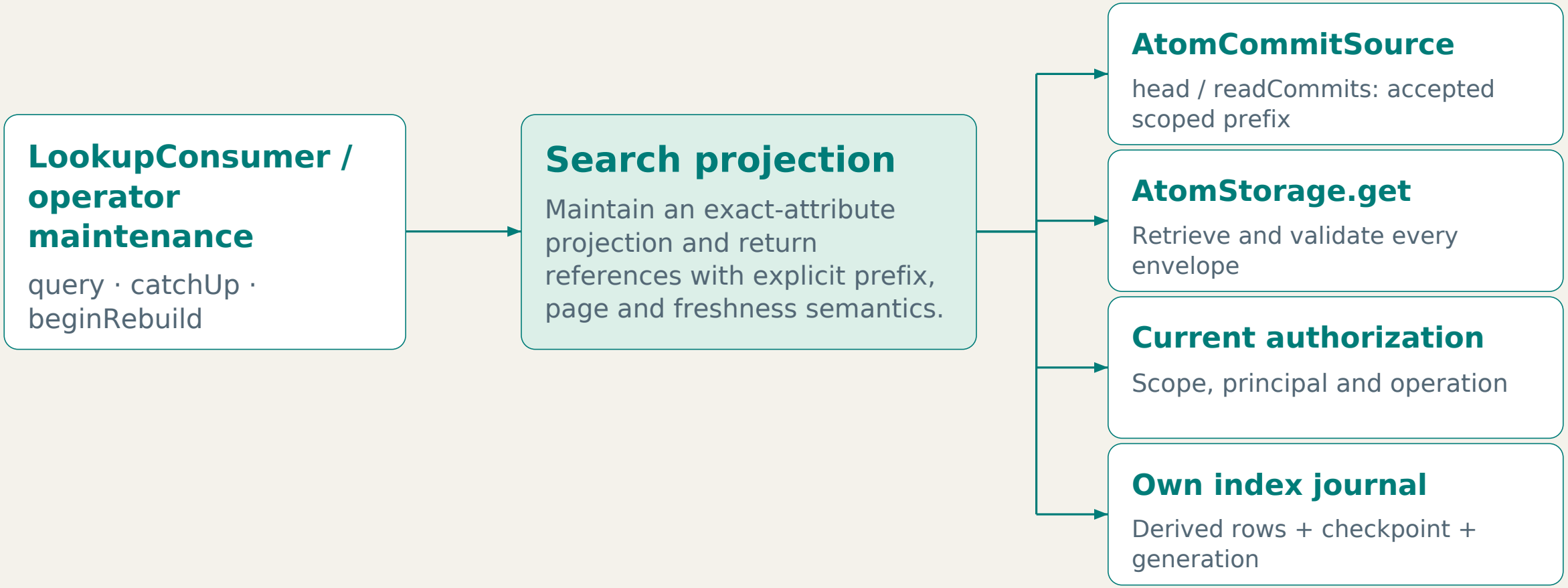
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	**Search projection**	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Maintain an exact-attribute projection and return references with explicit prefix, page and freshness semantics.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Derived projection

Extracted exact attributes; disposable and rebuildable

Progress and generation

Atomic index/checkpoint updates, rebuild intent and cursor generation

OUTSIDE THIS RESPONSIBILITY

Authoritative source

Acceptance and immutable bytes belong to AtomStorage

Semantic cognition

No embeddings, scoring, inferred tasks or general knowledge search

exact-attributes/1 · knowledge-attributes/1 · contract 2.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
query	Exact conjunction of 1–3 distinct attributes	kind/topic/label; equality only
catchUp	Process ≤8 new source entries	Checkpoint advances only after complete batch
beginRebuild	Capture source head; create generation	Queries unavailable until captured prefix complete

256 indexed records; page/batch ≤8; query ≤1024 bytes; no automatic repair.

A worked example through the boundary

REAL HIVE + SQLITE / DeTTa · independent file Search remains a fixture

01 · STEP

Initial discovery

Five matches, three pages at limit two.

02 · STEP

Append a match

Requested barrier exceeds the indexed prefix.

03 · STEP

Signal lag

query returns NOT_AVAILABLE; no ingestion.

04 · STEP

Catch up / rebuild

Six roots and nine payloads; replacement agrees.

What the example establishes

An answer is useful only when its freshness means something precise.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Exact matching

Case/Unicode preserved; missing attributes never match.

I2

Complete prefix

No advance across gaps, unavailable bytes or forged metadata.

I3

Bound continuation

Cursor fixes query, principal, source, snapshot and generation.

I4

Honest freshness

Prefix completeness, page completion, requested minimum and observed head are distinct.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Before/after checkpoint commit	Rows and checkpoint recover together.	Both providers; four direct kills
During rebuild intent	Resume bounded rebuild; reject old-generation cursor.	Both providers; two direct kills
Native page/result crash	Resume caller progress without seeded answers.	Both providers; four native kills

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

5 → 6

Matching roots before/after catch-up

Native discovery counts, not a relevance score

2

Independent Search providers

SQLite owner + atomic-file fixture

256 / 8

Indexed records / max page or batch

Finite contract limits, not scaling results

Native tests compare complete reference sets and canonical payload bytes across four transitions. No query-latency, ingestion-rate or relevance benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary corpus size, selectivity, page size and index lag; report p50/p95 query latency, catch-up rate and rebuild time with explicit freshness barriers.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	SQLite SearchOwner over accepted real DeTTa bytes; actual native Hive query path.
FIXTURE / LOCAL TEST POLICY	Independent atomic-file Search with its own predicate evaluator; generated corpus/oracle.
BLOCKED / UNSUPPORTED	No required M5 gate blocked; general search and migration are outside this profile.
UNRUN	Embedding relevance, large-corpus scale, distributed indexing, compaction and live migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure freshness/latency tradeoff	Frozen corpus, source barriers and controlled lag	Latency and staleness reported together
Broaden query language deliberately	New explicit profile and sound cursor semantics	No silent change to exact equality
Scale rebuild machinery	Capacity/retention contract before optimization	No skipped source ranges or query-time ingestion

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

An answer is useful only when its freshness means something precise.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --gates hive --environment <env> --output <fresh>
```

Requires pinned native providers. The writer exits and the oracle is removed before native lookup. Fresh replacement rebuilds derived state; it is not migration.

`fusion/search.py`
`fusion/adapters/fixture_search.py`
`fusion/lookup_contracts.py`
`tests/test_m5_boundaries.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.